From a single characterisation snapshot to end-of-life: a PyBaMM-trained neural operator for second-life LFP RUL

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Second-life reuse of used lithium-iron-phosphate (LFP) cells for stationary energy storage is bottlenecked by characterisation cost: per-cell PyBaMM [1] calibration of degradation parameters against measured state-of-health (SoH) trajectories requires $\gtrsim 400$ cycles per cell [2]. Prior second-life work [3] and universal-differential-equation degradation [4] improve calibration *quality* without shrinking the cycling budget. We ask: *can a physics-conditioned neural operator, pretrained on PyBaMM-generated trajectories, predict the full SoH curve of a new cell from a single characterisation snapshot plus a short cycling window?*

We construct a two-stage pipeline. *Stage 1 — Physics corpus:* starting from PyBaMM's Prada2013 LFP chemistry with per-cell BOL parameters identified from OCV/GITT/HPPC/RPT on a supplier-B fresh-cell cohort (OCV RMSE 6.8 mV median), we Sobol-sweep 300 simulations across five degradation channels (SEI kinetics, plating, $\mathrm{LAM_{pos}}$, $\mathrm{LAM_{neg}}$), producing 300 synthetic SoH trajectories with paired ground-truth θ. *Stage 2 — Neural operator:* a DeepONet [5] variant whose branch consumes a five-stream fingerprint (DCIR, RPT, first-K measured SoH, θ, protocol) and whose trunk consumes the target cycle number; output $\mathrm{SoH}(n) = \mathrm{SoH_{init}} - \mathrm{softplus}(\langle b, t\rangle)$ is hard-monotonic by construction. θ-conditioning lets the operator absorb electrochemistry *directly* from Phase-1 param-ID rather than reconstructing it from characterisation.

Trained on the 300-sim corpus with $K = 50$ input cycles, the operator predicts held-out synthetic trajectories to **1.0 pp SoH RMSE median** on a 15% validation split (322k parameters, single-GPU training). Applied to a deep-second-life supplier-A cell at $K = 50$, the model forecasts remaining useful life to a second-life EoSL threshold of SoH $= 0.40$ (below first-life $0.70{-}0.80$ because repurposing pack builders run cells at low C-rates where usefulness extends further) — a **1376-cycle RUL from 50 measured cycles** (Fig. 1). This decouples the cycling budget from the extrapolation horizon.

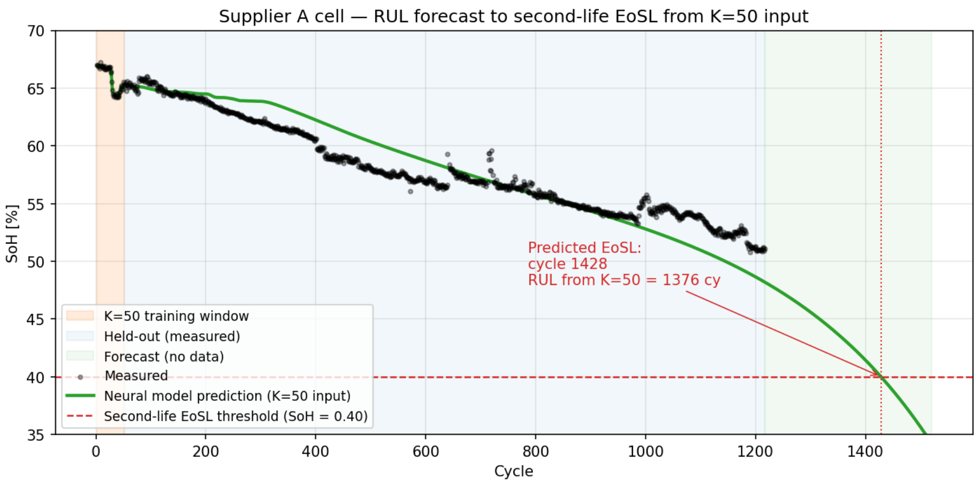

Figure 1: Neural-model RUL forecast for a deep-second-life supplier-A cell. From $K = 50$ measured cycles (orange band), the model predicts across the held-out measurement range (blue) and extrapolates past the last measurement (green) to cross the second-life EoSL threshold (SoH $= 0.40$) at cycle 1428, forecasting a total RUL of 1376 cycles. Baseline uses our warmstart PINN; the neural-operator formulation described here retains the K=50 input budget and extends output to the full remaining-life horizon via PyBaMM-generated training trajectories.

References. [1] V. Sulzer et al., *PyBaMM*, J. Open Res. Softw. 9(1), 14–21 (2021). [2] M. Prada et al., *Simplified electrochemical and thermal model of LiFePO$_4$-graphite Li-ion batteries*, J. Electrochem. Soc. 160(4), A616–A628 (2013). [3] O. Ouledboutaarija et al., *Advancing Second-Life Lithium-Ion Batteries*, PyBaMM Conf. 2025, p. 19. [4] J. A. Kuzhiyil et al., *Enhancing Generalisability of Physics-based Battery Degradation Using UDE*, PyBaMM Conf. 2025, p. 12. [5] L. Lu et al., *DeepONet: learning nonlinear operators for identifying differential equations*, Nat. Mach. Intell. 3, 218–229 (2021).